

Sparse Matrix Lesson 1 Practice Worksheet

Progress checkpoint

Current stage: Lesson 1 — Sparse Matrix Foundations.

Before moving to Lesson 2, you should be able to:

- Explain what a sparse matrix is.
- Identify `m`, `n`, and `num`.
- Convert a small matrix into Coordinate List form.
- Build the CSR arrays `A`, `IA`, and `JA`.
- Explain why `IA` has `rows + 1` entries.
- Compare full matrix storage with CSR storage.

Practice 1: Identify sparse data

Given:

```
0 0 8 0
0 0 0 0
5 0 0 2
```

Answer:

```
m =
n =
num =
```

Practice 2: Coordinate List

Convert the same matrix into Coordinate List form:

```
Row  Column  Value
```

With metadata row:

```
Row  Column  Value
```

Practice 3: CSR

Build:

A =
JA =
IA =

Practice 4: Explain in one sentence

Why does CSR save memory compared with a normal 2D matrix?

Answer:

Practice 5: Exit check

Explain why `IA` has `rows + 1` entries.

Answer: